



# CV2090

## Probability & Statistics for Civil Engineers

(Jul – Nov 2026)

Dr. Prakash S Badal

# Mean, Variance of Different Distributions

| Distr.            | Description/Example                                                                                                                                                   | $p_X(x), \mu, \sigma^2$                                                                                                               |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Bernoulli         | $x$<br>A coin toss                                                                                                                                                    | $p_X(x) = p^x(1-p)^{1-x}, x = 0,1$<br>$\mu = p, \sigma^2 = pq$                                                                        |
| Binomial          | exactly $x$ success in $n$ trials<br>Prob. of 4 Heads on 10 coin flips?                                                                                               | $p_X(x) = \binom{n}{x} p^x q^{n-x}$<br>$\mu = np, \sigma^2 = npq$                                                                     |
| Geometric         | $x$ trials required for the first success<br>Prob. of getting first H on 10th toss?                                                                                   | $p_X(x) = (1-p)^{x-1}p$<br>$\mu = \frac{1}{p}, \sigma^2 = \frac{(1-p)}{p^2}$                                                          |
| Poisson           | $x$ events in a fixed time interval, when events occur independently at a constant average rate.<br>Prob. of 3 accidents at an intersection in a day?                 | $p_X(x) = \frac{e^{-\lambda} \lambda^x}{x!}, x = 0,1,2, \dots$<br>$\mu = \lambda, \sigma^2 = \lambda$                                 |
| Negative Binomial | $x$ trials needed to obtain $r$ successes<br>Prob. of getting 4 <sup>th</sup> Head on 10th coin toss?                                                                 | $p_X(x) = \binom{x-1}{r-1} (1-p)^{x-r} p^r$<br>$\mu = \frac{r}{p}, \sigma^2 = \frac{r(1-p)}{p^2}$                                     |
| Hypergeometric    | $x$ success in $n$ trials, from a population of size $N$ , which has $K$ successes (without replacement)<br>Prob. of getting two Aces in 4 picks without replacement? | $p_X(x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}$<br>$\mu = np$ , where $p = K/N$<br>$\sigma^2 = np(1-p) \frac{N-n}{N-1}$ |

L12  
Aug 20, 2026

# Announcement

- 20<sup>th</sup> Aug: tut-3
- 21<sup>st</sup> Aug: Class tomorrow (Friday) at 10 AM
- 24<sup>th</sup> Aug: Lecture hour ~~today~~ 3 – 4 pm; tutorial 4 – 5 pm
- 25<sup>th</sup> Aug: No class on Tuesday
- 28<sup>th</sup> Aug: Quiz-1 on next Friday 8 AM



- Check course page on GitHub regularly

<https://github.com/prakashvs613/2026-CV2090-IITM/>

# Higher Moments

Expectation of a function of  $X$

$$\mu = E[X] = \sum_{x_i} x_i p_X(x_i) = \int_{-\infty}^{\infty} x f_X(x) dx$$

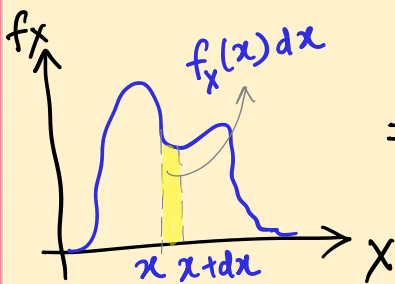
$$E[g(X)] = \int_{-\infty}^{\infty} g(x) \cdot f_X(x) \cdot dx$$

Let  $Y = g(X)$

$$E[Y] = \int_{-\infty}^{\infty} y \cdot f_Y(y) \cdot dy$$

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) \cdot f_{g(X)}(g(x)) \cdot dg(x)$$

Assume one-to-one  $g(X)$



# Expectation and Moments

$$\mu_{g(x)} = E[g(x)] = \int_{-\infty}^{\infty} g(x) f_x(x) dx$$

$$\mu = E[X] = \int_{-\infty}^{\infty} x f_x(x) dx$$

$$\sigma^2 = E[(X-\mu)^2] = \int_{-\infty}^{\infty} (x-\mu)^2 f_x(x) dx = E[X^2] - \mu^2$$

$$m^{\text{th}} \text{ moment of } X = E[X^m]$$

$$1^{\text{st}} \text{ central moment of } X = E[(X-\mu)]$$

$$2^{\text{nd}} \text{ central moment of } X = E[(X-\mu)^2] = \sigma^2$$

$$\int_{-\infty}^{\infty} x^2 f_x(x) dx \rightarrow 2^{\text{nd}} \text{ moment of } X$$

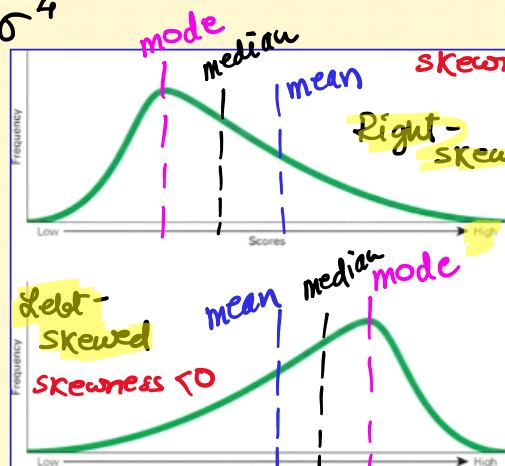
## Central Moments, Skewness

$$1^{\text{st}} \text{ central moment } E[X-\mu] = 0$$

$$2^{\text{nd}} \text{ central moment } E[(X-\mu)^2] = \sigma^2$$

$$3^{\text{rd}} \text{ central moment } E[(X-\mu)^3] = \text{skewness}$$

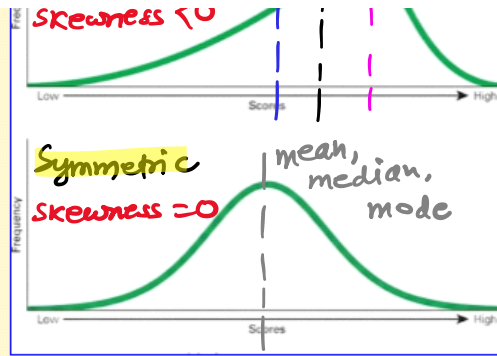
$$4^{\text{th}} \text{ central moment } \frac{E[(X-\mu)^4]}{\sigma^4} = \text{kurtosis}$$



Qualitatively mark

- Mean,
- Median, and
- Mode,

on all three figures.



on all three figures.

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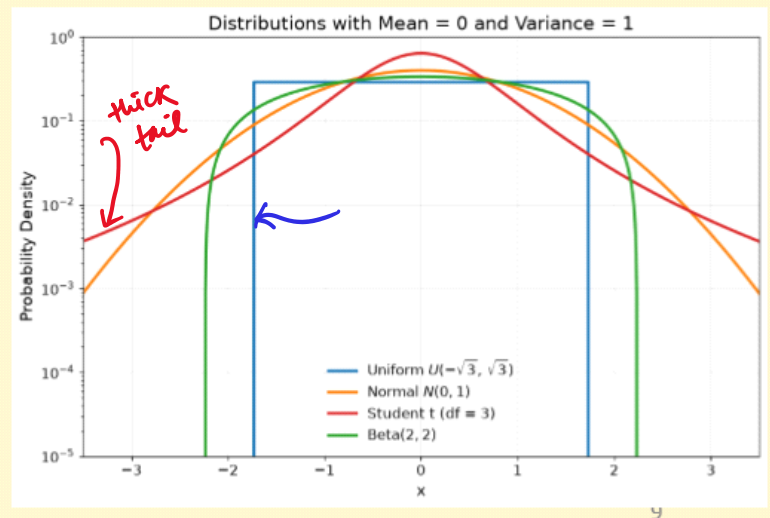
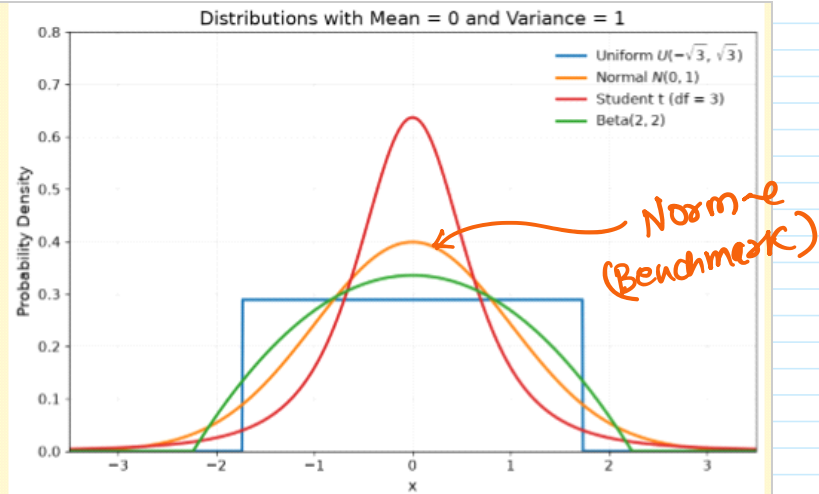
# Kurtosis

Excess kurtosis

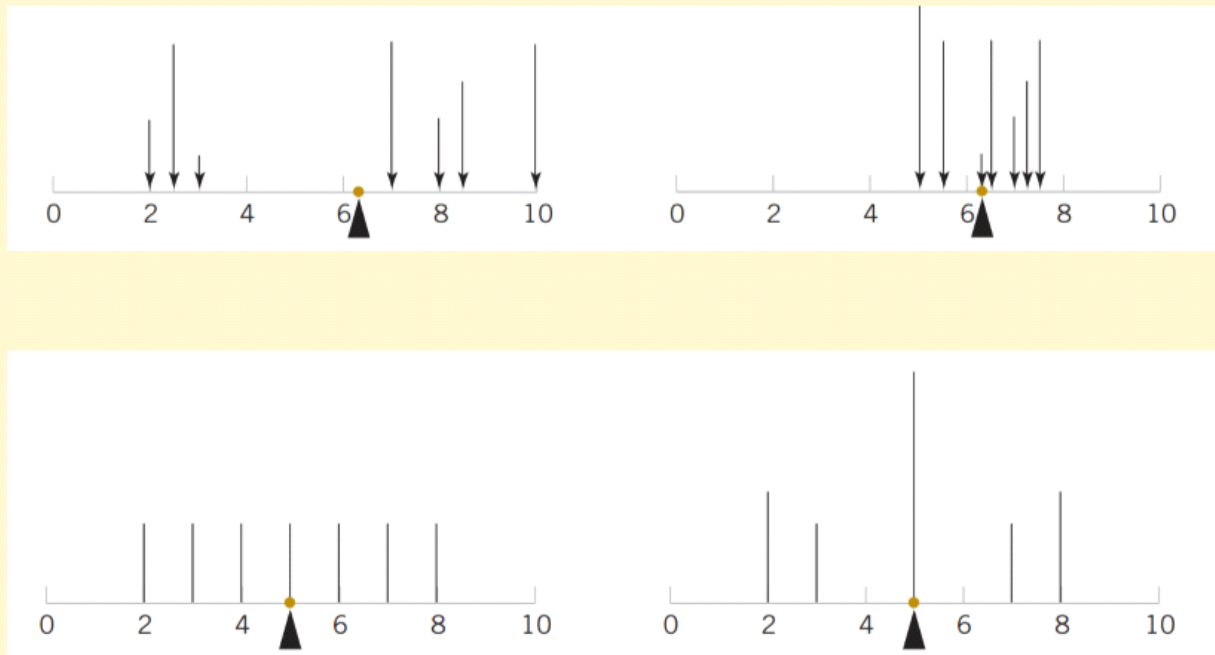
$$\frac{E[(X-\mu)^4]}{\sigma^4} - 3$$

↑  
Benchmark

Excess Kurtosis  $> 0$ , thick/heavy tail  
 $< 0$ , light tail  
 $= 0$ , Normal



$\{\text{Mean, Variance}\} \neq \text{PMF/PDF}$



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# Tyrannosaurus Rex (T-Rex)

Rex = King



# Tyrannosaurus Rex (T-Rex)

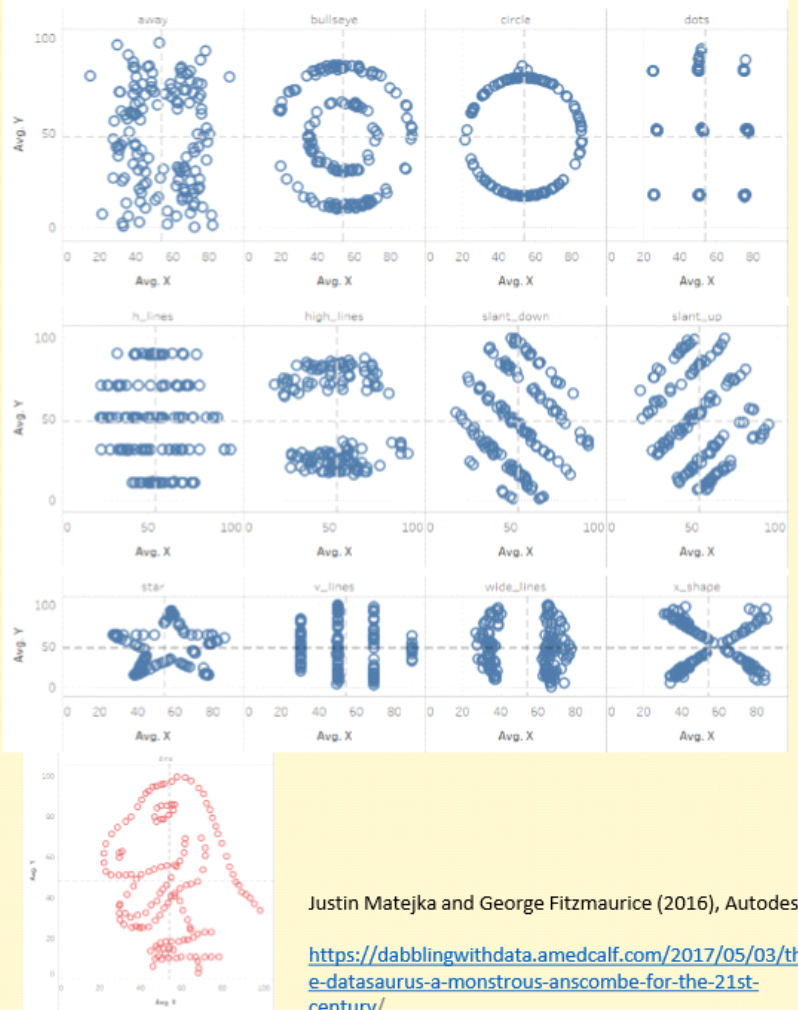


Thick tail

# Datasaurus Dozen

Baker's dozen

Datasaurus Dozen



Justin Matejka and George Fitzmaurice (2016), Autodesk

<https://dabblingwithdata.amedcalf.com/2017/05/03/the-datasaurus-a-monstrous-anscombe-for-the-21st-century/>

# Datasaurus Dozen

<https://seanfobbe.com/tutorials/importance-of-visualization-datasaurus/>

|     | dataset    | x_mean   | x_sd     | y_mean   | y_sd     | corr        |
|-----|------------|----------|----------|----------|----------|-------------|
|     | <char>     | <num>    | <num>    | <num>    | <num>    | <num>       |
| 1:  | dino       | 54.26327 | 16.76514 | 47.83225 | 26.93540 | -0.06447185 |
| 2:  | away       | 54.26610 | 16.76982 | 47.83472 | 26.93974 | -0.06412835 |
| 3:  | h_lines    | 54.26144 | 16.76590 | 47.83025 | 26.93988 | -0.06171484 |
| 4:  | v_lines    | 54.26993 | 16.76996 | 47.83699 | 26.93768 | -0.06944557 |
| 5:  | x_shape    | 54.26015 | 16.76996 | 47.83972 | 26.93000 | -0.06558334 |
| 6:  | star       | 54.26734 | 16.76896 | 47.83955 | 26.93027 | -0.06296110 |
| 7:  | high_lines | 54.26881 | 16.76670 | 47.83545 | 26.94000 | -0.06850422 |
| 8:  | dots       | 54.26030 | 16.76774 | 47.83983 | 26.93019 | -0.06034144 |
| 9:  | circle     | 54.26732 | 16.76001 | 47.83772 | 26.93004 | -0.06834336 |
| 10: | bullseye   | 54.26873 | 16.76924 | 47.83082 | 26.93573 | -0.06858639 |
| 11: | slant_up   | 54.26588 | 16.76885 | 47.83150 | 26.93861 | -0.06860921 |
| 12: | slant_down | 54.26785 | 16.76676 | 47.83590 | 26.93610 | -0.06897974 |
| 13: | wide_lines | 54.26692 | 16.77000 | 47.83160 | 26.93790 | -0.06657523 |

$$\{\mu, \sigma^2\} \neq \text{PMF/PDF}$$

Datasaurus Dozen

